

Airline Network Planning & Route Optimization | Business Use Case

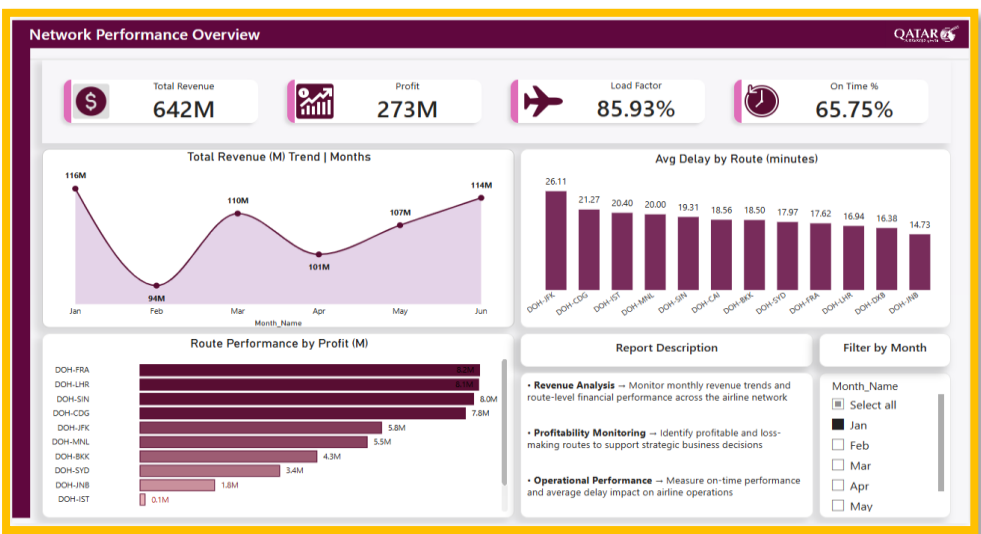
Position Reference: Senior Corporate Planning Analyst – Qatar Airways

Project Objective: This project was inspired by the Qatar Airways Network Planning role, focused on transforming operational, financial, supply, and demand airline data into actionable business insights for strategic decision-making.

Key Responsibilities from Job Description: Monitor operational KPIs and profitability, support network planning and schedule optimization, analyse operational and financial datasets, build dashboards for executive decision-making & support route optimization and capacity planning

Proposed Solution: An interactive Power BI analytical solution was designed to support:

- Route profitability analysis
- Operational KPI monitoring
- Capacity vs demand analysis
- Executive performance reporting
- Strategic route optimization



Core Business Problems

Problem 1: Route Profitability Imbalance

Profitability imbalance identified between passenger demand and route-level operational cost.

Problem 2: Operational Delay Issues

Operational delays reduce airline efficiency and negatively impact performance KPIs.

Problem 3: Capacity vs Demand Imbalance

Passenger demand and aircraft capacity may not properly align across routes.

Proposed Analytical Solution

Tools Used

- Power BI
- Power Query
- DAX
- Data Modelling

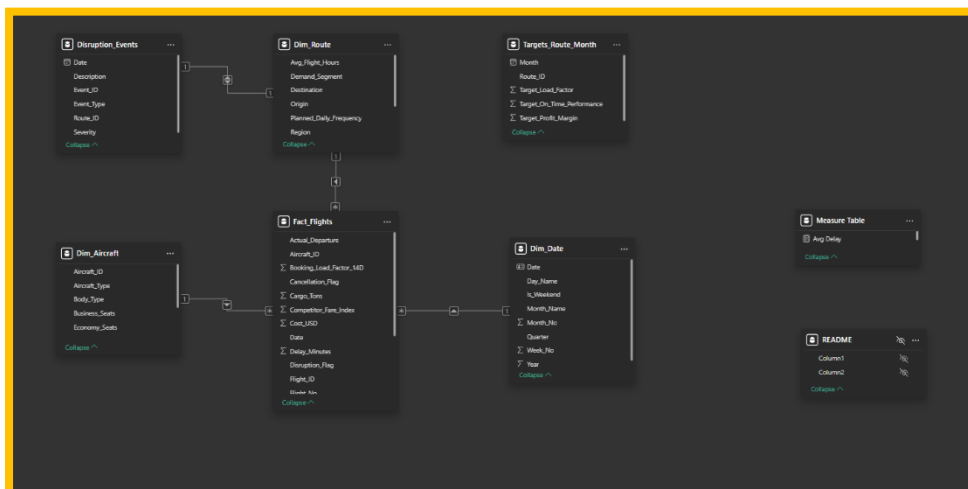
Dataset

Airline operational dataset generated using AI tools for simulation purposes.

Solution Components

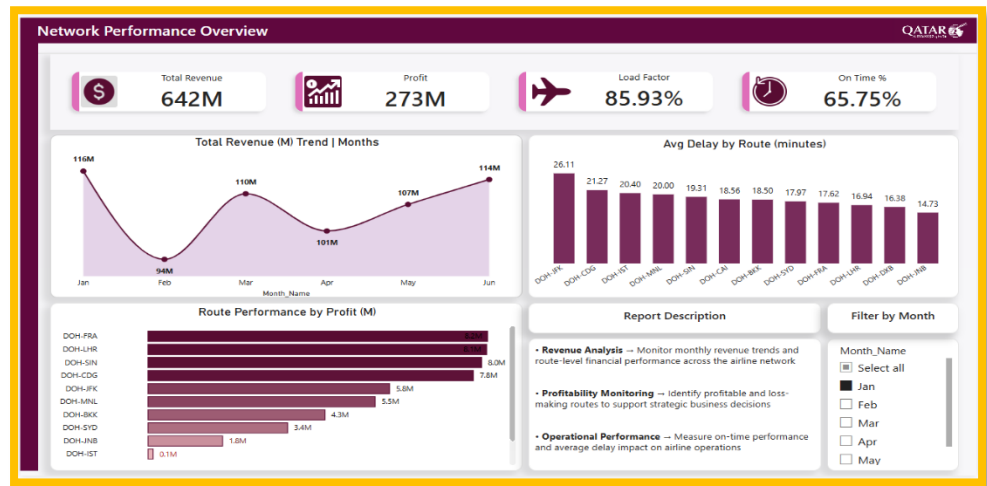
- **Network Performance Overview:** Executive-level operational and financial KPI monitoring.
- **Route Optimization Analysis:** Profitability and route optimization analysis.
- **Selected Route Analysis:** Detailed operational and profitability breakdown for selected routes.

Data Model



Dashboard Pages

Network Performance Overview



Problem 1 → Solution from Dashboard

Problem: Profitability imbalance identified between passenger demand and operational cost across airline routes.

Analytical Approach

The report analyses: Route profitability, Revenue vs operating cost, Demand vs profitability imbalance & Loss-making routes

Supporting Visuals

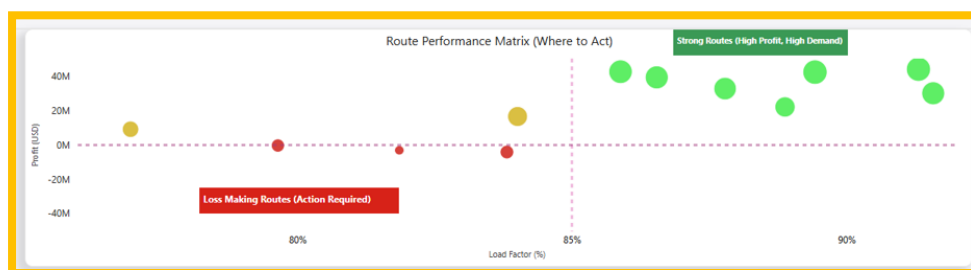
1. Route Optimization Matrix
2. Top Loss Routes
3. Revenue vs Operating Cost
4. Route Performance Table

Business Value

Supports: Route optimization, Pricing review, Cost reduction decisions, Strategic commercial planning

Key Insights

- Strong routes indicate expansion opportunities.
- Loss-making routes require operational review.
- High demand but weak profitability may indicate cost inefficiencies.



Problem 2 → Solution from Dashboard

Problem: Operational delays reduce airline efficiency and affect route-level operational performance.

Analytical Approach

The report analyses: Average delay by route, Operational KPIs, Route-level performance trends & Delay-heavy routes

Supporting Visuals

1. Avg Delay by Route
2. Network Performance Overview
3. Selected Route Analysis

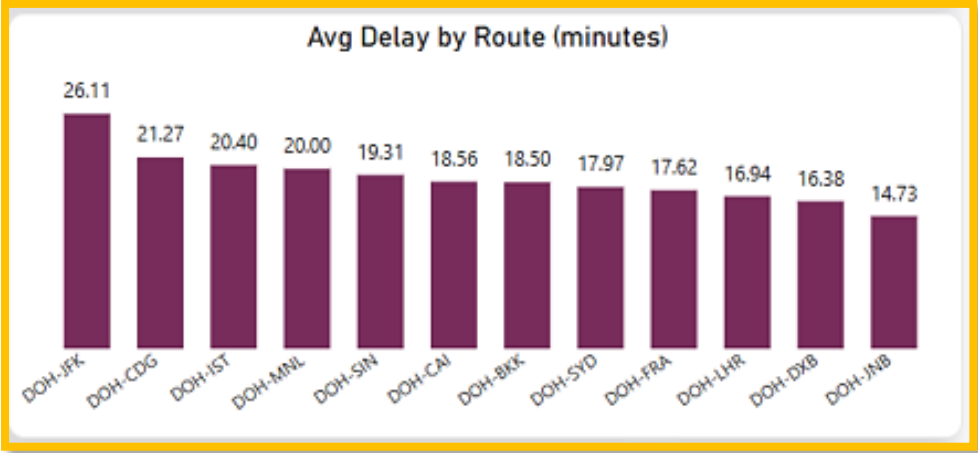
Business Value

Supports:

- Delay reduction strategies
- Schedule optimization
- Operational KPI monitoring
- Performance improvement

Key Insights

- Delay-heavy routes can be identified quickly.



- Operational KPIs help monitor route efficiency.
- Selected route analysis supports operational investigation.



Problem 3 → Solution from Dashboard

Problem: Passenger demand and aircraft capacity may not be properly balanced across airline routes.

Analytical Approach

The report analyses: Capacity vs demand, Passenger utilization, Load factor trends & Underutilized routes

Supporting Visuals

1. Capacity vs Demand Analysis
2. Load Factor KPIs
3. Passenger Utilization Metrics
4. Route Performance Table

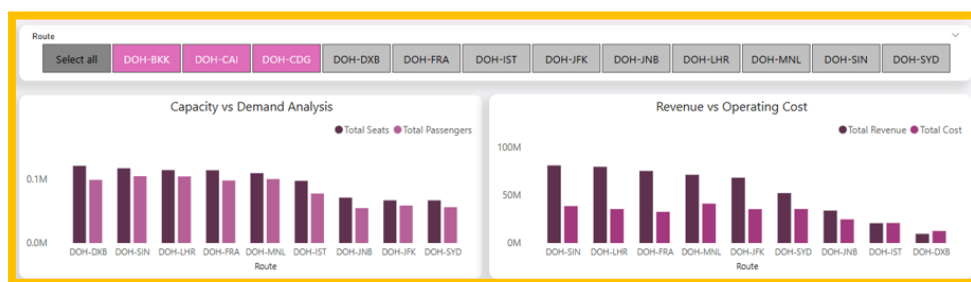
Business Value

Supports:

- Capacity planning
- Aircraft allocation optimization
- Improved seat utilization
- Demand planning

Key Insights

- Passenger demand can be compared against seat capacity.
- Load factor monitoring supports capacity optimization.
- Underutilized routes can be identified for operational review.



Final Summary

Business Impact

The solution helps support: Route optimization, Operational KPI monitoring, Capacity planning, Profitability analysis & Executive decision-making



Final Statement

This project demonstrates how Power BI and analytical reporting can support airline network planning and operational decision-making inspired by the Qatar Airways Senior Corporate Planning Analyst role.

The solution transforms operational and financial airline data into actionable business insights through interactive dashboards and analytical reporting.